20261022 Quiz

for (MATH261) Discrete Mathematics

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Problem 1. Solve the following recurrence relations subject to the given initial conditions.

(a) For $n \geq 2$,

$$a_n = 6a_{n-1} - 8a_{n-2}, \qquad a_0 = 1, \quad a_1 = 0.$$

(b) For $n \geq 2$,

$$2a_n = 7a_{n-1} - 3a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

Solution.

(a) The characteristic equation is

$$r^2 - 6r + 8 = (r-2)(r-4) = 0.$$

Since the roots 2 and 4 are distinct, the general solution is

$$a_n = \alpha 2^n + \beta 4^n.$$

The initial conditions give

$$\alpha + \beta = 1, \qquad 2\alpha + 4\beta = 0,$$

whence $\alpha = 2$ and $\beta = -1$. Therefore

$$a_n = 2^{n+1} - 4^n.$$

(b) Substituting $a_n = r^n$ gives the characteristic equation

$$2r^2 - 7r + 3 = (2r-1)(r-3) = 0.$$

Its distinct roots are 3 and $\frac{1}{2}$, so

$$a_n = \alpha 3^n + \beta \left(\frac{1}{2}\right)^n.$$

From $a_0 = 1$ and $a_1 = 1$,

$$\alpha + \beta = 1, \qquad 3\alpha + \frac{\beta}{2} = 1.$$

Thus $\alpha = \frac{1}{5}$ and $\beta = \frac{4}{5}$, giving

$$a_n = \frac{3^n + 4(1/2)^n}{5}.$$

□

Problem 2. Solve the following recurrence relations subject to the given initial conditions.

(a) For $n \geq 2$,

$$a_n = -2na_{n-1} + 3n(n-1)a_{n-2}, \qquad a_0 = 1, \quad a_1 = 2.$$

(b) For $n \geq 2$,

$$c_n = 2 + \sum_{i=1}^{n-1} c_i, \qquad c_1 = 1.$$

Solution.

(a) Define $b_n = a_n/n!$ for $n \geq 0$. Dividing the recurrence by $n!$ and using $n! = n(n-1)!$ yields

$$\begin{aligned} b_n &= -2\frac{a_{n-1}}{(n-1)!} + 3\frac{a_{n-2}}{(n-2)!} \\ &= -2b_{n-1} + 3b_{n-2}, \qquad n \geq 2. \end{aligned}$$

The initial values are $b_0 = 1$ and $b_1 = 2$. The characteristic equation is

$$r^2 + 2r - 3 = (r-1)(r+3) = 0,$$

so $b_n = \alpha + \beta(-3)^n$. Applying the initial conditions gives

$$\alpha + \beta = 1, \qquad \alpha - 3\beta = 2,$$

and hence $\alpha = \frac{5}{4}$, $\beta = -\frac{1}{4}$. Returning to $a_n = n!b_n$, we obtain

$$a_n = \frac{n!}{4}(5 - (-3)^n).$$

(b) The equation at $n = 2$ gives

$$c_2 = 2 + c_1 = 3.$$

For $n \geq 3$, subtract the recurrence at $n-1$ from the one at n:

$$\begin{aligned} c_n - c_{n-1} &= \left(2 + \sum_{i=1}^{n-1} c_i\right) - \left(2 + \sum_{i=1}^{n-2} c_i\right) \\ &= c_{n-1}. \end{aligned}$$

Thus $c_n = 2c_{n-1}$ for $n \geq 3$. Together with $c_1 = 1$ and $c_2 = 3$, this gives

$$c_1 = 1, \qquad c_n = 3 \cdot 2^{n-2} \quad (n \geq 2).$$

□

Problem 3. Find the general solutions of the following recurrence relations.

(a) For $n \geq 2$,
$$2a_n = 7a_{n-1} - 3a_{n-2} + 2^n.$$

(b) For $n \geq 2$,
$$9a_n = 6a_{n-1} - a_{n-2} + 5n^2.$$

Solution.

(a) The associated homogeneous recurrence has characteristic equation
$$2r^2 - 7r + 3 = (2r-1)(r-3) = 0,$$
so its general solution is $a_n^{(h)} = \alpha 3^n + \beta(1/2)^n$. Since 2 is not a characteristic root, seek a particular solution of the form $a_n^{(p)} = C2^n$. Substitution gives
$$2C = \frac{7C}{2} - \frac{3C}{4} + 1, \qquad C = -\frac{4}{3}.$$
Therefore the general solution is
$$a_n = \alpha 3^n + \beta\left(\frac{1}{2}\right)^n - \frac{4}{3}\,2^n,$$
where $\alpha, \beta \in \mathbb{R}$ are arbitrary.

(b) The characteristic equation of the associated homogeneous recurrence is
$$9r^2 - 6r + 1 = (3r-1)^2 = 0.$$
The root $\frac{1}{3}$ has multiplicity two, so
$$a_n^{(h)} = (\alpha + \beta n)\left(\frac{1}{3}\right)^n.$$
Seek a polynomial particular solution $a_n^{(p)} = An^2 + Bn + C$. Substitution into
$$9a_n^{(p)} - 6a_{n-1}^{(p)} + a_{n-2}^{(p)} = 5n^2$$
and expansion give
$$4An^2 + (8A + 4B)n + (-2A + 4B + 4C) = 5n^2.$$
Comparing coefficients yields
$$4A = 5, \qquad 8A + 4B = 0, \qquad -2A + 4B + 4C = 0,$$
hence $A = \frac{5}{4}$, $B = -\frac{5}{2}$, and $C = \frac{25}{8}$. Thus
$$a_n = (\alpha + \beta n)\left(\frac{1}{3}\right)^n + \frac{5}{4}n^2 - \frac{5}{2}n + \frac{25}{8},$$
where $\alpha, \beta \in \mathbb{R}$ are arbitrary.

□

Problem 4. Find a solution of the linear nonhomogeneous recurrence relation

$$a_{n+3} - 7a_{n+2} + 16a_{n+1} - 12a_n = n4^n, \qquad n \geq 0,$$

subject to $a_0 = -2$, $a_1 = 0$, and $a_2 = 5$.

Solution. The characteristic polynomial of the associated homogeneous recurrence factors as

$$r^3 - 7r^2 + 16r - 12 = (r-2)^2(r-3).$$

Thus its general solution is

$$a_n^{(h)} = (\alpha + \beta n)2^n + \gamma 3^n.$$

Since 4 is not a characteristic root and the nonhomogeneous term is $n4^n$, seek a particular solution $a_n^{(p)} = (An + B)4^n$. Substitution gives

$$\begin{aligned} a_{n+3}^{(p)} &- 7a_{n+2}^{(p)} + 16a_{n+1}^{(p)} - 12a_n^{(p)} \\ &= 4^n\big[64(A(n+3)+B) - 112(A(n+2)+B) \\ &\qquad + 64(A(n+1)+B) - 12(An+B)\big] \\ &= 4^n(4An + 32A + 4B). \end{aligned}$$

Matching this with $n4^n$ gives

$$4A = 1, \qquad 32A + 4B = 0, \qquad \text{so} \qquad A = \frac{1}{4}, \quad B = -2.$$

Consequently,

$$a_n = (\alpha + \beta n)2^n + \gamma 3^n + \left(\frac{n}{4} - 2\right)4^n.$$

Applying the three initial conditions yields

$$\begin{aligned} \alpha + \gamma &= 0, \\ 2\alpha + 2\beta + 3\gamma &= 7, \\ 4\alpha + 8\beta + 9\gamma &= 29. \end{aligned}$$

Using $\gamma = -\alpha$, the remaining equations reduce to

$$2\beta - \alpha = 7, \qquad 8\beta - 5\alpha = 29.$$

Thus $\alpha = -1$, $\beta = 3$, and $\gamma = 1$. Therefore

$$a_n = (3n-1)2^n + 3^n + \left(\frac{n}{4} - 2\right)4^n.$$

□